Governing Equations of the **fuzzy-couscous** Solver

Clean-room compressible LES solver for blast-induced turbulence

Generated 2026-05-23

Abstract

This document collects every governing equation the **fuzzy-couscous** solver discretizes: the compressible Navier–Stokes system, the inviscid and viscous fluxes, the equations of state (ideal gas, two-γ mixture, Jones–Wilkins–Lee, and stiffened gas), the multifluid models (the transported-γ marker and the five-equation diffuse-interface model), the LES closure (hyperdissipation, localized artificial diffusivity, Ducros sensor), the Besnard–Harlow–Rauenzahn (BHR) variable-density turbulence model, the SSP-RK3 time integrator, and the diagnostic identities. Each section cites the file that implements it, so the equations stay traceable to the code.

Contents

1 Compressible Navier–Stokes system

The solver advances the conservative compressible Navier–Stokes equations for a single fluid (or a mixture; see §4) on a periodic / slip-wall Cartesian box. With conserved state $\mathbf{U} = (\rho,\ \rho\mathbf{u},\ \rho E)^{\mathsf{T}}$,

$$\frac{\partial \rho}{\partial t} + \nabla\cdot(\rho\mathbf{u}) = 0, \tag{1}$$

$$\frac{\partial(\rho\mathbf{u})}{\partial t} + \nabla\cdot(\rho\mathbf{u}\otimes\mathbf{u} + p\,\mathbf{I}) = \nabla\cdot\boldsymbol{\tau}, \tag{2}$$

$$\frac{\partial(\rho E)}{\partial t} + \nabla\cdot[(\rho E + p)\,\mathbf{u}] = \nabla\cdot(\boldsymbol{\tau}\cdot\mathbf{u}) - \nabla\cdot\mathbf{q}, \tag{3}$$

where ρ is density, $\mathbf{u} = (u, v, w)$ velocity, p pressure, and the total specific energy is $E = e + \frac{1}{2}|\mathbf{u}|^2$ with e the specific internal energy. The right-hand side is assembled as a telescoping difference of face fluxes, $R_i \mathrel{-}= (F_{i+1/2} - F_{i-1/2})/\Delta x$, so the discrete scheme is conservative by construction (`numerics/RHS.cpp`).

2 Inviscid flux

The hyperbolic (Euler) flux in direction $d \in \{x, y, z\}$ with unit normal $\mathbf{n}_d$ and normal velocity $u_d = \mathbf{u}\cdot\mathbf{n}_d$ is

$$\mathbf{F}_d^{\mathrm{euler}}(\mathbf{U}, p) = \begin{pmatrix} \rho u_d \\ \rho u\,u_d + p\,n_{d,x} \\ \rho v\,u_d + p\,n_{d,y} \\ \rho w\,u_d + p\,n_{d,z} \\ (\rho E + p)\,u_d \end{pmatrix}. \tag{4}$$

It is evaluated by a hybrid 6th-order central / 5th-order WENO reconstruction [4, 10] in flux-difference form with Lax–Friedrichs splitting; the WENO weight is gated by the shock sensor of §6 (`physics/EulerFlux.hpp`, `numerics/WENO5.hpp`). The flux is EOS-agnostic: p is supplied by §4.

3 Viscous fluxes

The viscous stress is the compressible Stokes tensor with an optional bulk viscosity μ_b,

$$\tau_{ij} = \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \tfrac{2}{3}\,\delta_{ij}\,\nabla\cdot\mathbf{u}\right) + \mu_b\,\delta_{ij}\,\nabla\cdot\mathbf{u}, \tag{5}$$

and the heat flux is Fourier's law

$$\mathbf{q} = -\kappa\,\nabla T, \qquad \kappa = \frac{\mu\,c_p}{\mathrm{Pr}}, \tag{6}$$

with T the temperature and Pr the Prandtl number. First derivatives use composed 6th-order central stencils (NGHOST $= 6$), so the viscous Laplacian is 6th-order accurate (`physics/ViscousFlux.hpp`, `numerics/Gradients.cpp`).

4 Equation of state

The pressure closure maps (marker, ρ, e) $\mapsto (p, c)$ through a single mixture dispatcher (`physics/MixtureEOS.hpp`), where c is the sound speed used at the flux and CFL sites.

4.1 Ideal gas

$$p = (\gamma - 1)\,\rho e, \qquad c^2 = \frac{\gamma p}{\rho}. \tag{7}$$

4.2 Two-γ mixture (variable-density contact)

A per-cell advected marker $G = 1/(\gamma - 1)$ selects between ambient air and a second ideal gas (products with γ_p). The effective ratio of specific heats is recovered as $\gamma = 1 + 1/G$, giving

$$p = \frac{\rho e}{G}, \qquad c^2 = \frac{(1 + 1/G)\,p}{\rho}. \tag{8}$$

This is the original variable-density contact model (`physics/Multifluid.hpp`).

4.3 Jones–Wilkins–Lee (JWL) for real high explosives

Detonation products (e.g. TNT) obey the JWL EOS, which an ideal gas cannot represent. With relative volume $V = \rho_0/\rho$ (ρ_0 the reference/solid density) and Grüneisen coefficient ω,

$$p(\rho, e) = A\left(1 - \frac{\omega}{R_1 V}\right) e^{-R_1 V} + B\left(1 - \frac{\omega}{R_2 V}\right) e^{-R_2 V} + \omega\,\rho\,e. \tag{9}$$

Writing $f(V)$ for the reference-curve (first two) terms, $p = f(V) + \omega\rho e$, which inverts in closed form for the internal energy at fixed density, $e = (p - f(V))/(\omega\rho)$, and yields the isentropic sound speed $c^2 = (\partial p/\partial\rho)_s$. With $A = B = 0$ the EOS reduces to an ideal gas with $\gamma = 1 + \omega$, the hook that lets the mixture dispatcher treat air and products uniformly (`physics/JWL.hpp`).

The shipped TNT parameter set (LLNL / Dobratz & Crawford [20]) is:

Parameter	Symbol	Value
Reference coefficient	A	371.2 GPa
Reference coefficient	B	3.231 GPa
Reference exponent	R_1	4.15
Reference exponent	R_2	0.95
Grüneisen coefficient	ω	0.30
Reference density	ρ_0	1630 $\mathrm{kg/m^3}$
Detonation energy	E_0	7.0 GPa
CJ pressure	p_{CJ}	21 GPa
Detonation speed	D	6930 m/s

Runs are nondimensionalized with $\rho_{\mathrm{ref}} = \rho_0$ and $p_{\mathrm{ref}} = p_{\mathrm{CJ}}$ so the $\sim$1000:1 density and $\sim 10^5$:1 pressure contrasts (Atwood ≈ 0.999) stay well conditioned. The CJ density $\rho_{\mathrm{CJ}} = \rho_0/(1 - p_{\mathrm{CJ}}/(\rho_0 D^2))$ is the Rankine–Hugoniot value, self-consistent with the JWL sound speed at CJ ($c_{\mathrm{CJ}} = D - u_{\mathrm{CJ}}$).

4.4 Stiffened gas

The stiffened-gas (Noble–Abel/Tammann) EOS adds a constant stiffness pressure p_∞ to the ideal-gas law so that nearly incompressible phases (water, condensed explosives) share the same Grüneisen-type closure as a gas:

$$p = (\gamma - 1)\,\rho e - \gamma\, p_\infty, \qquad c^2 = \frac{\gamma\,(p + p_\infty)}{\rho}. \tag{10}$$

With $p_\infty = 0$ this reduces to the ideal gas (the analogue of JWL's $A = B = 0$ reduction), so a stiffened-gas phase doubles as an ideal-gas phase (`physics/StiffenedGas.hpp`).

4.5 Five-equation mixture (volume-fraction / Wood rule)

For the five-equation model (§5.3) [11, 12] two phases coexist in each cell with volume fraction α_1 ($\alpha_2 = 1 - \alpha_1$) and partial densities $Z_k = \alpha_k \rho_k$. Each phase obeys a Grüneisen split of its internal-energy density, $\rho_k e_k = (p + \Phi_k)/(\Gamma_k - 1)$, where

$$\Gamma_k - 1 = \begin{cases} \gamma_k - 1 & \text{stiffened gas} \\ \omega_k & \text{JWL} \end{cases}, \qquad \Phi_k = \begin{cases} \gamma_k\, p_{\infty,k} & \text{stiffened gas} \\ -f_k(V_k) & \text{JWL (ref. curve, §4.3)} \end{cases}. \tag{11}$$

Summing $\rho e = \sum_k \alpha_k \rho_k e_k$ over the phases gives the volume-fraction-averaged (Wood) pressure

$$p = \frac{\rho e - \Pi}{S}, \qquad S = \sum_k \frac{\alpha_k}{\Gamma_k - 1}, \qquad \Pi = \sum_k \frac{\alpha_k\, \Phi_k}{\Gamma_k - 1}, \tag{12}$$

with ρe the mixture internal-energy density and $\rho = Z_1 + Z_2$. The frozen mixture sound speed is the mass-weighted phase average

$$c^2 = \sum_k Y_k\, c_k^2, \qquad Y_k = Z_k/\rho, \tag{13}$$

where c_k^2 is the per-phase stiffened-gas/JWL sound speed at the common pressure p. For stiffened-gas phases S and Π are linear in α_1 — the property that lets an isolated interface stay pressure-oscillation-free (`physics/MixtureEOS.hpp`, `p_c_5eq`).

5 Multifluid models

5.1 Model taxonomy

The solver carries two multifluid models, both single-velocity and single-pressure, related by the diffuse-interface hierarchy (Baer–Nunziato's seven-equation non-equilibrium system [1] being the full parent):

- **Transported-γ four-equation model** [5, 9, 2, 3]: the three conservative hydrodynamic equations of §1 plus *one* advected EOS marker (below). The equation of state is selected per cell by the marker; there are no per-phase masses. This is distinct from the *mass-fraction* four-equation model (two partial masses closed by temperature equilibrium [12]), which this code does not implement.

- **Five-equation diffuse-interface model** [11, 8]: velocity- and pressure-equilibrium two-phase flow with two partial-mass equations, mixture momentum and energy, and a volume-fraction transport (§5.3).

The $4 \to 5$ transition splits the single mixture mass into the two partial masses $\alpha_1\rho_1, \alpha_2\rho_2$ and replaces the transported γ marker by the volume fraction α_1.

5.2 Transported-γ marker transport

The marker (the two-γ field G, or the products mass fraction $\phi \in [0, 1]$ for JWL) is advected with the flow,

$$\frac{\partial(\rho\,\chi)}{\partial t} + \nabla\cdot(\rho\,\chi\,\mathbf{u}) = 0, \qquad \chi \in \{G,\ \phi\}, \tag{14}$$

equivalently $\partial\chi/\partial t + \mathbf{u}\cdot\nabla\chi = 0$ by continuity. The contact is handled either by a gated double-flux (non-oscillatory) or an opt-in fully conservative flux (`conservative = true`); the latter conserves total energy at the cost of small contact oscillations (`physics/Multifluid.cpp`). The double-flux method — freezing the per-cell EOS across the flux stencil so a single-material conservative scheme handles the contact without spurious pressure oscillation — is that of Abgrall & Karni [9] (building on Karni [2] and Abgrall [3]); "gated" here denotes that it is applied only within a few cells of a contact (bulk cells take the cheap shared flux), an implementation optimization rather than a separately named scheme.

5.3 Five-equation diffuse-interface model

The true five-equation system [11, 8] advances the two partial masses, the mixture momentum and energy, and the volume fraction:

$$\frac{\partial(\alpha_1\rho_1)}{\partial t} + \nabla\cdot(\alpha_1\rho_1\,\mathbf{u}) = 0, \tag{15}$$

$$\frac{\partial(\alpha_2\rho_2)}{\partial t} + \nabla\cdot(\alpha_2\rho_2\,\mathbf{u}) = 0, \tag{16}$$

$$\frac{\partial(\rho\mathbf{u})}{\partial t} + \nabla\cdot(\rho\mathbf{u}\otimes\mathbf{u} + p\,\mathbf{I}) = \nabla\cdot\boldsymbol{\tau}, \tag{17}$$

$$\frac{\partial(\rho E)}{\partial t} + \nabla\cdot[(\rho E + p)\,\mathbf{u}] = \nabla\cdot(\boldsymbol{\tau}\cdot\mathbf{u}) - \nabla\cdot\mathbf{q}, \tag{18}$$

$$\frac{\partial\alpha_1}{\partial t} + \mathbf{u}\cdot\nabla\alpha_1 = 0, \tag{19}$$

with mixture density $\rho = \alpha_1\rho_1 + \alpha_2\rho_2$ and pressure p from the Wood-rule law (§4.5). The non-conservative volume-fraction equation is discretized in quasi-conservative form, $\partial\alpha_1/\partial t + \nabla\cdot(\alpha_1\mathbf{u}) = \alpha_1\,\nabla\cdot\mathbf{u}$, using the same face velocities as the conservative fluxes. The two partial-mass fluxes reuse the same reconstruction as the mixture-mass flux, so $Z_1 + Z_2 \equiv \rho$ to round-off, and the interface stays pressure/velocity-equilibrium (no spurious oscillation at a contact). The state is advanced in lockstep with the conserved fields through every SSP-RK3 stage (`numerics/RHS.cpp`, `add_inviscid_divergence_5eq`; `numerics/RK3.cpp`, `step_5eq/step_mpi_5eq`).

Shock capturing for the five-equation system. All five equations — conservative momentum and energy, the two conservative partial masses (with $\rho = Z_1 + Z_2$ derived), and the non-conservative volume fraction α_1 — are advanced through the *same* shock-capturing engine: one Ducros sensor θ (§6, dilated ± 2 cells along the face direction), one hybrid reconstruction that uses 6th-order central differencing where $\theta \le 0.65$ and 5th-order WENO where $\theta > 0.65$, and one Lax–Friedrichs split with a *single shared* wave speed $\alpha = |u_d| + c$ (c the frozen mixture sound speed). Because every conserved and non-conserved quantity sees the same α and the same sensor, the contact remains in pressure/velocity equilibrium. Unlike the two-γ model (§5.2) the five-equation model therefore needs *no* double-flux: the stiffened-gas mixture closure makes the closure coefficients S and Π *linear* in α_1 (§4.5), so the volume-fraction-averaged pressure $p = (\rho e - \Pi)/S$ stays uniform across a uniform-$(p, \mathbf{u})$ interface — the sensor sees no jump, WENO is not triggered, and the plain conservative flux is already oscillation-free (`numerics/RHS.cpp`, `fill_flux_alpha_5eq` / `add_inviscid_divergence_5eq`).

6 LES closure and shock capturing

6.1 Spectral hyperdissipation (LES sink)

The sub-grid sink is a hyper-viscous term applied to every conserved variable,

$$\left.\frac{\partial \mathbf{U}}{\partial t}\right|_{\mathrm{LES}} = -\,\nu_h\,\nabla^4\mathbf{U}, \tag{20}$$

whose discrete operator matches the analytic k^4 damping; under SSP-RK3 it preserves the $e^{-\nu_h k^4 t}$ eigenvalue decay (`numerics/HyperdissipationSpectral.cpp`). The empirical stability invariant is fixed $\nu_{\mathrm{eff}} = \nu_h k_{\max}^2$; between resolutions ν_h scales as Δx^2 (see the README TGV note).

6.2 Ducros shock sensor

The hybrid central/WENO blend [10] and the artificial diffusivity are gated by the Ducros sensor [7]

$$\theta = \frac{(\nabla\cdot\mathbf{u})^2}{(\nabla\cdot\mathbf{u})^2 + |\boldsymbol{\omega}|^2 + \varepsilon}, \tag{21}$$

with vorticity $\boldsymbol{\omega} = \nabla\times\mathbf{u}$ and a small ε; $\theta\to 1$ in compression (shocks), $\theta \to 0$ in smooth/vortical regions. A pressure-jump tanh ramp and a half-width-2 dilation along the face direction sharpen the gate (`numerics/Ducros.cpp`).

6.3 Localized artificial diffusivity (LAD)

For strong blasts the Cook/Kawai–Lele localized artificial diffusivity [14, 15] adds grid-dependent, sensor-localized dissipation to shocks and contacts,

$$\mu^* = C_\mu\,\rho\,|\nabla^r(\nabla\cdot\mathbf{u})|\,\Delta^{r+2}, \qquad \beta^* = C_\beta\,\rho\,|\nabla^r(\nabla\cdot\mathbf{u})|\,\Delta^{r+2}, \tag{22}$$

with an analogous artificial mass/contact diffusivity D^* for the density-contact piece; the canonical order is $r = 4$ (`numerics/ArtificialDiffusivity.cpp`).

6.4 Artificial diffusivity for the five-equation interface

The base LAD above acts on the conserved state only. For the five-equation model (§5.3) it is extended to the aux fields so the material interface is regularized *without* breaking the contact's pressure/velocity equilibrium (`numerics/ArtificialDiffusivity.cpp`, `compute_lad_fields_5eq`; `numerics/RHS.cpp` `add_rhs_viscous_5eq`). Two changes are required. First, the contact diffusivity is re-keyed on the *interface* markers rather than the mixture density (which can be smooth where α_1 jumps),

$$D^* = C_D\,c_{\mathrm{mix}}\,\max\!\Big(|\nabla^r\alpha_1|,\ \tfrac{|\nabla^r Z_1|}{Z_1},\ \tfrac{|\nabla^r Z_2|}{Z_2}\Big)\,\Delta, \tag{23}$$

with c_{mix} the mixture sound speed (consistent with the inviscid wave speed $\alpha = |u_d| + c$). Second, the thermal term is disabled ($\kappa^* = 0$): the cell-gradient temperature is an ideal-gas reduction that is *non-uniform* across a uniform-pressure contact, so a thermal flux keyed on it would inject spurious pressure; the bulk/shear terms are retained (they vanish at a stationary contact).

The interface is then regularized by a consistent diffusion [17]: the two partial masses, the volume fraction, *and* the internal-energy density $\rho e_{\text{int}} = \rho E - \frac{1}{2}\rho|\mathbf{u}|^2$ all diffuse with the *same* coefficient and stencil,

$$\partial_t q \mathrel{+}= \nabla\cdot(D^*\nabla q), \qquad q \in \{Z_1, Z_2, \alpha_1, \rho e_{\text{int}}\}, \tag{24}$$

with the mixture-mass flux taken as the sum of the partial-mass fluxes (so $\rho = Z_1 + Z_2$ is preserved) and momentum carried at the local velocity. Because the stiffened-gas closure makes $\rho e_{\text{int}} = p\,S(\alpha_1) + \Pi(\alpha_1)$ *affine* in α_1 (§4.5), applying the same linear operator to α_1 and to ρe_{int} leaves the state on the manifold $p = (\rho e_{\text{int}} - \Pi)/S$ at the *unchanged* p: a uniform-$(p, \mathbf{u})$ interface stays in pressure equilibrium to round-off (exact for stiffened-gas phases; approximate for JWL phases, where Π is non-linear in the phase density). This is verified end-to-end by `tests/test_five_eq_abv.cpp` (stationary contact: p uniform to 10^{-9} while α_1 spreads; shock crossing an interface with LAD active stays bounded and positive).

6.5 Robustness: closely-spaced shocks and positivity

WENO5-JS [4] assumes *at most one* discontinuity inside its five-point stencil, so that at least one of its three candidate sub-stencils is smooth. Two shocks that approach within roughly a stencil width violate this premise: all three smoothness indicators $\beta_0, \beta_1, \beta_2$ become large and comparable, the nonlinear weights can no longer isolate a smooth sub-stencil, and the scheme loses both its design order and its non-oscillatory property. In the worst case a strong expansion trapped between the two shocks drives the reconstructed density or internal energy negative, which makes the sound speed $c = \sqrt{\gamma p/\rho}$ undefined and poisons the field. The reconstruction here is plain WENO5-JS rather than the more robust WENO-Z [16] or mapped WENO [13], so this regime is guarded by several explicit mechanisms rather than by the reconstruction alone:

1. **Sensor dilation.** The ± 2-cell max-dilation of θ (§6) forces WENO over the *entire* region spanning both shocks, so the central scheme never rings in the gap between them.

2. **Lax–Friedrichs dissipation.** The shocked-face flux carries local LF dissipation with $\alpha = |u_d| + c$, taken as the max over the two adjacent cells; LF is the most dissipative (most robust) split and damps the inter-shock oscillations the reconstruction may generate.

3. **Sound-speed guards.** Every $c = \sqrt{\cdot}$ is taken as $\sqrt{\max(\cdot, 0)}$ (`physics/MixtureEOS.hpp`), so a transient negative c^2 cannot inject a NaN before the floors below act.

4. **Per-stage positivity floors.** After *each* of the three SSP-RK3 stages the state is repaired before the next sound-speed evaluation: `enforce_positivity` floors ρ and the internal energy (guaranteeing $p > 0$ independent of the local γ, `core/State.hpp`), and for the five-equation model `enforce_5eq_bounds` clamps $\alpha_1 \in [\alpha_{\text{floor}}, 1 - \alpha_{\text{floor}}]$, keeps $Z_1, Z_2 \geq z_{\text{floor}}$, and resets $\rho = Z_1 + Z_2$ (`physics/Multifluid.cpp`). The floors are auto-scaled to the ambient state so resolved physics is untouched; the per-stage clamp count is reported as a diagnostic.

5. **LAD backstop.** The localized artificial diffusivity above is an opt-in, tunable dissipation sink that targets exactly the shock and contact cells (with a `disable_weno` "LAD-only" mode); its diffusivity also enters the viscous time-step limit, so added robustness stays time-stable.

These are floors and added dissipation, not a provably positivity-preserving limiter [18]; the reconstruction is also component-wise rather than characteristic-wise. In practice the combination keeps shock–shock and shock–contact interactions bounded (exercised by the Shu–Osher and five-equation shock-bubble tests), at the cost of extra local dissipation where shocks crowd together.

7 BHR variable-density turbulence model

The Besnard–Harlow–Rauenzahn [19] k–ε–a–b model runs as an operator-split sub-step alongside the LES hydro. The six primitives are the turbulent kinetic energy k, its dissipation ε, the turbulent mass flux a_i, and the density–specific-volume covariance $b = -\overline{\rho' \, (1/\rho)'}$. The dominant TKE source is the variable-density (baroclinic) production

$$P_{\mathrm{VD}} = -\, a_i \, \frac{\partial p}{\partial x_i}, \tag{25}$$

validated as dominant in the LES a/b budget. The turbulent viscosity is $\nu_t = C_\mu \, k^2/\varepsilon$, and the modeled transport equations carry gradient-diffusion terms with Schmidt numbers $(\sigma_k, \sigma_\varepsilon, \sigma_a, \sigma_b)$ (`turbulence/BHR.hpp`, `turbulence/BHR.cpp`).

When two-way feedback is enabled, the model returns a Reynolds-stress divergence and dissipative heating to $\mathbf{U}$, blended by the resolution-adequacy sensor

$$f_k = \frac{k_{\mathrm{sgs}}}{k_{\mathrm{res}} + k_{\mathrm{sgs}}}, \qquad k_{\mathrm{res}} = \tfrac{1}{2} \left| \mathbf{u} - \widetilde{\mathbf{u}} \right|^2, \tag{26}$$

where $\widetilde{\mathbf{u}}$ is a test-filtered velocity (resolved energy in the $[\Delta x, 2\Delta x]$ octave); $f_k \to 1$ under-resolved (RANS), $f_k \to 0$ when the LES resolves the energy.

8 Time integration

Time advance uses the three-stage strong-stability-preserving Runge–Kutta (SSP-RK3, Gottlieb–Shu [6]) scheme on $\mathbf{U}^n \to \mathbf{U}^{n+1}$ with right-hand side $\mathcal{L}(\mathbf{U})$:

$$\mathbf{U}^{(1)} = \mathbf{U}^n + \Delta t \, \mathcal{L}(\mathbf{U}^n), \tag{27}$$

$$\mathbf{U}^{(2)} = \tfrac{3}{4} \, \mathbf{U}^n + \tfrac{1}{4} \, \mathbf{U}^{(1)} + \tfrac{1}{4} \, \Delta t \, \mathcal{L}(\mathbf{U}^{(1)}), \tag{28}$$

$$\mathbf{U}^{n+1} = \tfrac{1}{3} \, \mathbf{U}^n + \tfrac{2}{3} \, \mathbf{U}^{(2)} + \tfrac{2}{3} \, \Delta t \, \mathcal{L}(\mathbf{U}^{(2)}). \tag{29}$$

The time step obeys the minimum of a hyperbolic CFL ($\propto \Delta x/(|\mathbf{u}| + c)$) and a viscous limit; in MPI runs both are reduced with `MPI_Allreduce(MIN)` (`numerics/RK3.cpp`, `numerics/RHS.cpp`). Each of the three stages is followed by positivity/bounds enforcement (`enforce_positivity`, and for the five-equation model `enforce_5eq_bounds`); this per-stage repair is the primary robustness backstop for strong or closely-spaced shocks (§6.5).

9 Diagnostic identities

9.1 Helmholtz decomposition

The velocity is split into solenoidal and dilatational parts, $\mathbf{u} = \mathbf{u}_s + \mathbf{u}_d$ with $\nabla \cdot \mathbf{u}_s = 0$ and $\nabla \times \mathbf{u}_d = \mathbf{0}$, giving the kinetic-energy split $K = K_{\mathrm{sol}} + K_{\mathrm{dil}}$ and the per-shell spectra $E(k) = E_{\mathrm{sol}}(k) + E_{\mathrm{dil}}(k)$ (`diagnostics/Spectra.cpp`).

9.2 Dissipation budget

The resolved dissipation is split into solenoidal and dilatational contributions,

$$\varepsilon_{\mathrm{total}} = \frac{\mu}{\rho} \left\langle \tau_{ij} \, \partial_j u_i \right\rangle, \qquad \varepsilon_{\mathrm{sol}} = \nu \left\langle |\boldsymbol{\omega}|^2 \right\rangle, \qquad \varepsilon_{\mathrm{dil}} = \tfrac{4}{3} \, \nu \left\langle (\nabla \cdot \mathbf{u})^2 \right\rangle, \tag{30}$$

together with the turbulent Mach number $M_t = u_{\mathrm{rms}}/\langle c \rangle$ (`diagnostics/Statistics.cpp`).

9.3 Conservation monitors

For a closed chamber the totals E/E_0 and M/M_0 are conserved to round-off by the discrete divergence theorem (telescoping face fluxes plus $\mathbf{u} \cdot \mathbf{n} = 0$ at slip walls); the internal energy $e_{\text{int}} = \sum(\rho E - \frac{1}{2}\rho|\mathbf{u}|^2)$ is accumulated directly so $KE + e_{\text{int}} = e_{\text{total}}$ is a genuine consistency check rather than an identity (`diagnostics/Statistics.cpp`; see the README for the conservation-vs-truncation discussion).

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